

Blast Radius in Software Development

1. Definition

Blast radius refers to the scope of impact a failure, bug, or outage has on a system. It describes how much of the infrastructure, application, or user base is affected when something goes wrong.

2. Examples

- Small blast radius: A feature flag bug affects only a limited subset of users.
- Large blast radius: A shared database outage causes all API requests to fail.
- LLM system example: Without retries or rate limiting, traffic spikes can cause system-wide failure.

3. Why Blast Radius Matters

- Prevents cascading failures.
- Improves system reliability.
- Protects user experience and revenue.
- Enables faster recovery from incidents.

4. How to Reduce Blast Radius

- Use feature flags and gradual rollouts.
- Implement retries with exponential backoff and jitter.
- Add circuit breakers and rate limiting.
- Isolate services to avoid shared failure points.
- Use caching to reduce dependency load.

5. Summary

Blast radius measures how far a failure spreads across a system. Well-designed systems minimize blast radius through isolation, controlled rollouts, resilience patterns, and operational safeguards.